

Predicting Road Network Failure Probabilities for Post-Disaster Drone Routing

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1. Introduction & Motivation

In post-disaster scenarios, such as major earthquakes, rapidly assessing network connectivity and dispatching emergency services is critical. Unmanned Aerial Vehicles (UAVs or drones) are increasingly used for this purpose. However, optimizing the travel time and routing of these drones requires accurate predictions of which road segments (arcs) are likely to be blocked by debris.

Current mathematical optimization models often rely on assumed or randomly generated probabilistic information for arc failures. This project bridges the gap between theoretical optimization and real-world infrastructure by developing a data-driven model. By utilizing open-source geographic data, this study transforms raw infrastructure layouts into empirical vulnerability scores, providing realistic inputs that can significantly enhance the accuracy of post-disaster drone routing strategies.

2. Data Collection & Preprocessing

To fulfill the project's data enrichment requirement, a hybrid dataset was constructed using spatial data techniques:

- **Primary Data:** The foundational road network geometries (nodes and edges) for the selected study area were programmatically extracted from OpenStreetMap (OSM) utilizing the **OSMnx** Python library.
- **Data Enrichment:** To assess the risk of debris blocking a road, the primary dataset was enriched with building footprint data, also extracted via OSM.
- **Preprocessing:** Using **GeoPandas**, spatial joins were performed to map buildings to their nearest road segments. This allowed for the calculation of two critical new features: **building_count** (the absolute number of structures near a road) and **structural_density** (a normalized metric reflecting building concentration relative to road length). Missing values were handled, and isolated network nodes were pruned to ensure a strongly connected graph.

3. Exploratory Data Analysis & Hypothesis Testing

A comprehensive Exploratory Data Analysis (EDA) was conducted to understand the spatial and statistical distribution of the network.

- **Visual Findings:** Histograms revealed that both road length and building counts are heavily right-skewed, indicating a network dominated by short, local streets. Spatial visualization highlighted that areas of high structural density are highly localized rather than evenly distributed, forming specific "choke points" that are critical for drone routing.
- **Statistical Validation:** To ensure the engineered features were statistically sound, three hypothesis tests were conducted (Significance Level $\alpha = 0.05$):
 1. **Pearson Correlation:** Tested the relationship between road length and building count. The test revealed an extremely weak correlation, proving that physical road length alone is not a sufficient indicator of building density or potential debris volume.
 2. **Independent T-Test:** Compared the structural density of "Short" versus "Long" roads. The results rejected the null hypothesis ($p < 0.05$), confirming a significant difference in risk profiles between local streets and main arterials.
 3. **One-Way ANOVA:** Roads were categorized into Low, Medium, and High building tiers. The ANOVA test yielded a significant result ($p < 0.05$), validating that the calculated **structural_density** metric robustly differentiates between varying urban topologies.

4. Machine Learning Methodology

With the features validated, the project employed both unsupervised and supervised machine learning techniques to predict road vulnerability.

- **Unsupervised Learning (K-Means Clustering):** The road network was segmented into three distinct clusters based on structural properties. This successfully categorized the network into High, Medium, and Low-Risk profiles, allowing for immediate prioritization in emergency algorithms.
- **Supervised Learning (Random Forest Regressor):** To provide a continuous failure probability ("p" value) for routing algorithms, a Random Forest model was trained to predict a simulated vulnerability score. Random Forest was selected due to its robustness against outliers and its ability to capture non-linear, complex spatial relationships without requiring extensive parameter tuning.

5. Results & Discussion

The Machine Learning pipeline yielded strong predictive performance:

- **Model Accuracy:** The Random Forest Regressor achieved an **R^2 score of ~0.72**. This indicates that the model successfully captured the underlying variance and patterns within the spatial dataset.

- **Feature Importance:** The analysis revealed that `building_count` (51%) and `structural_density` (38%) were the strongest predictors of road vulnerability, while `road_length` accounted for only ~10%.
- **Implications:** These results strongly validate the initial project hypothesis: local building concentration is a vastly superior metric for predicting post-disaster road blockages compared to basic geographic features. The continuous vulnerability scores generated by this model can now be directly exported as empirical parameters into linear programming solvers (e.g., Gurobi) to solve the Vehicle Routing Problem (VRP) for drones.

6. Conclusion & Future Work

This project successfully demonstrated how public geographic data can be enriched and modeled to solve a critical humanitarian logistics problem. By validating the structural density metric through rigorous hypothesis testing and achieving a 0.72 R^2 score via Random Forest, the project achieved its goal of generating realistic road failure probabilities.

Future Work: The model's accuracy could be further enhanced by integrating Peak Ground Acceleration (PGA) seismic risk maps from AFAD or municipal data portals. Additionally, incorporating real-time traffic data or historical fault-line proximities would allow the vulnerability scores to become highly dynamic, adapting to specific earthquake scenarios.

7. AI Usage Declaration

In accordance with the DSA 210 academic integrity guidelines, I declare the use of AI tools (Large Language Models) during this project. AI was utilized strictly as an assistive tool for:

1. Generating boilerplate `matplotlib` and `seaborn` code for data visualization during the EDA phase.
 2. Formatting statistical hypothesis structures (H_0 and H_A) and syntax for `scipy.stats`.
 3. Assisting in the structural outlining and grammar refinement of this final report.
- All core project logic, data enrichment methodologies, interpretation of statistical results, and final model architectures were independently directed, verified, and concluded by the student.